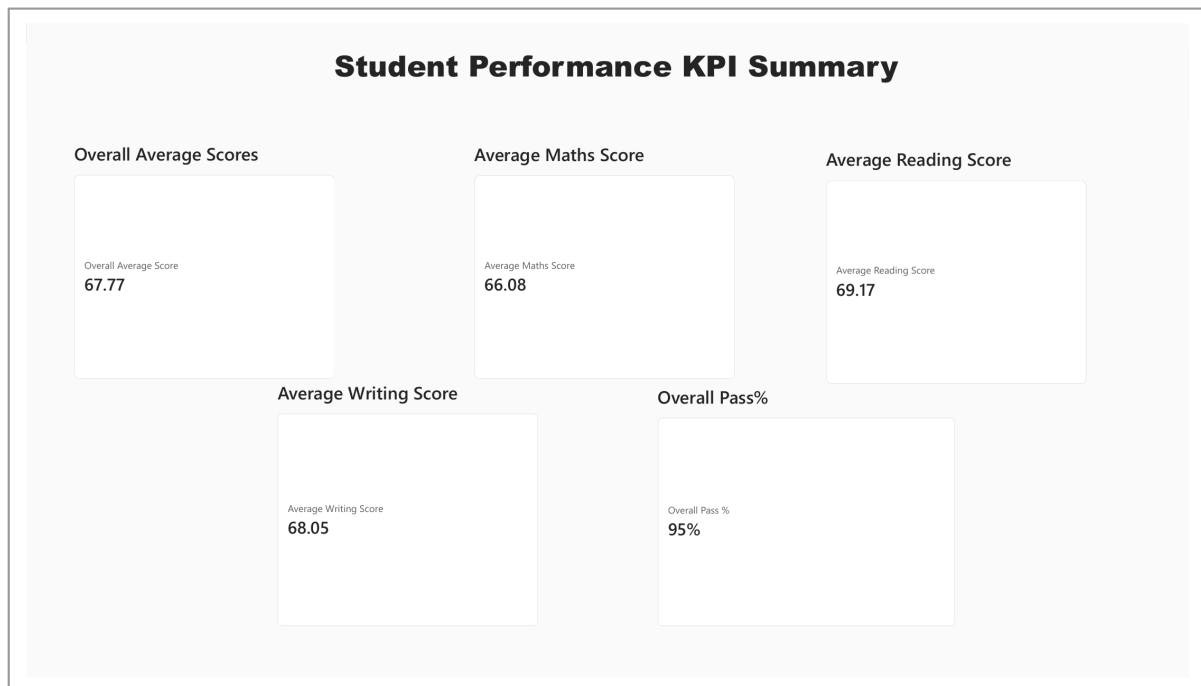


Week 8 Project Report

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1. Student Performance KPI Summary



This is the first report page in Power BI Desktop. It is titled Student Performance KPI Summary and uses five card visuals, each showing one headline number calculated from the student data. It lets the reader see the overall result without reading any charts.

Overall Average Scores (67.77): the mean score across all subjects and students, giving a single measure of general performance.

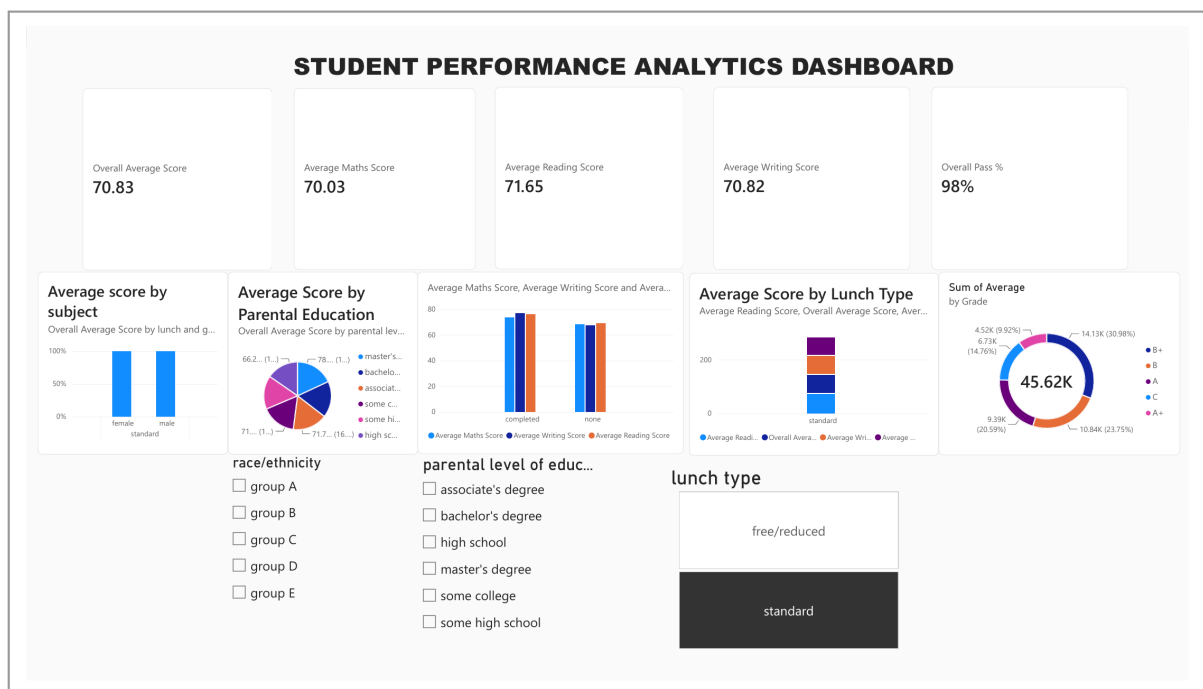
Average Maths Score (66.08): the mean marks in Mathematics. It is the lowest of the three subject averages.

Average Reading Score (69.17): the mean marks in Reading. It is the highest of the three subject averages.

Average Writing Score (68.05): the mean marks in Writing, which sits between Maths and Reading.

Overall Pass % (95%): the share of students who passed. A value this high shows that almost all students cleared the pass mark.

2. Student Performance Analytics Dashboard

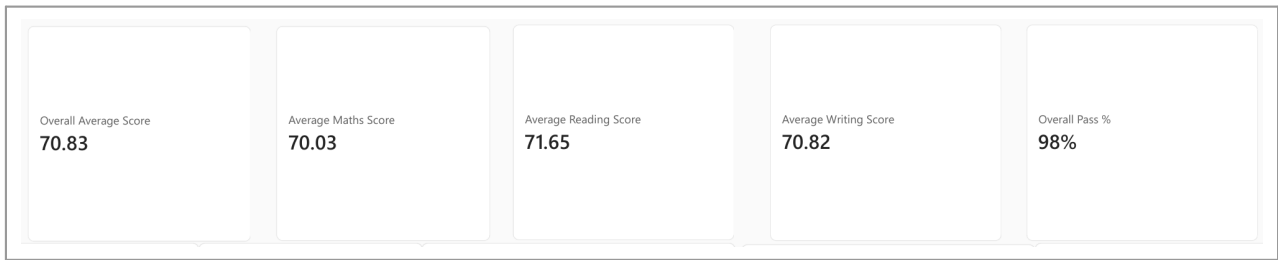


The second page is the main interactive dashboard, titled STUDENT PERFORMANCE ANALYTICS DASHBOARD. It combines KPI cards, five charts and three slicers on one page so that scores can be compared across gender, parental education, test preparation, lunch type and grade.

The page is laid out in three sections. The top row holds the KPI cards. The middle row holds the charts that break the scores down by different categories. The bottom row holds the slicers (race/ethnicity, parental level of education and lunch type), which filter every visual on the page when a value is selected.

The following pages explain each part of this dashboard separately.

2.1 KPI Cards

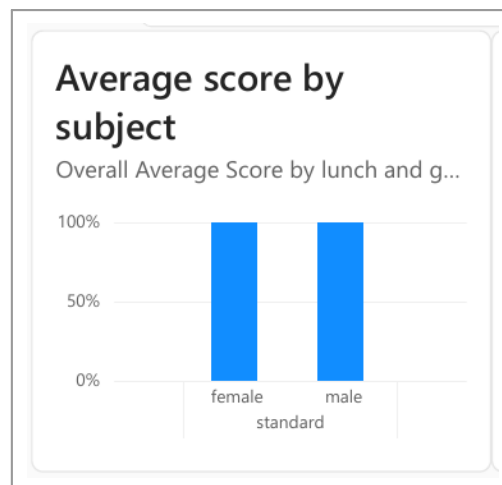


The top row has five cards: Overall Average Score (70.83), Average Maths Score (70.03), Average Reading Score (71.65), Average Writing Score (70.82) and Overall Pass % (98%).

These cards are built from DAX measures and work as the summary of the dashboard. Reading is again the strongest subject and Maths the weakest, but the gap between subjects is small, at roughly 1.6 marks. The pass percentage of 98% shows very strong results overall.

The values differ from the first KPI page (67.77 overall and 95% pass) because this dashboard is a separate page and its numbers can change with the filters or slicers applied. Because the cards sit on the same page as the slicers, they update whenever a slicer value is selected.

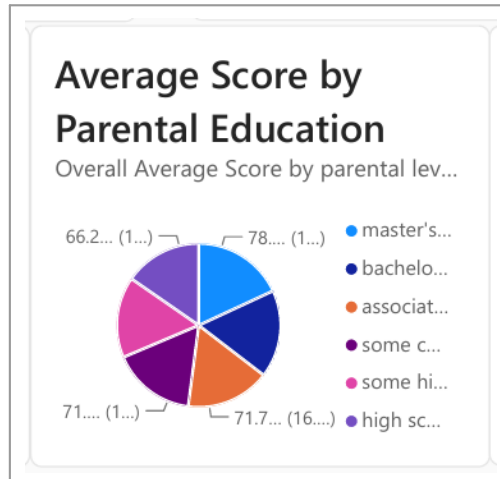
2.2 Average Score by Subject



This is a 100% stacked column chart with the subtitle Overall Average Score by lunch and gender. It has two columns, female and male, both under the lunch category standard.

Only one measure, the Overall Average Score, is plotted, so each column fills 100%. The chart therefore does not show a difference between the two genders. It confirms that both are represented under the standard lunch type. A clustered column chart would show the actual score values more clearly.

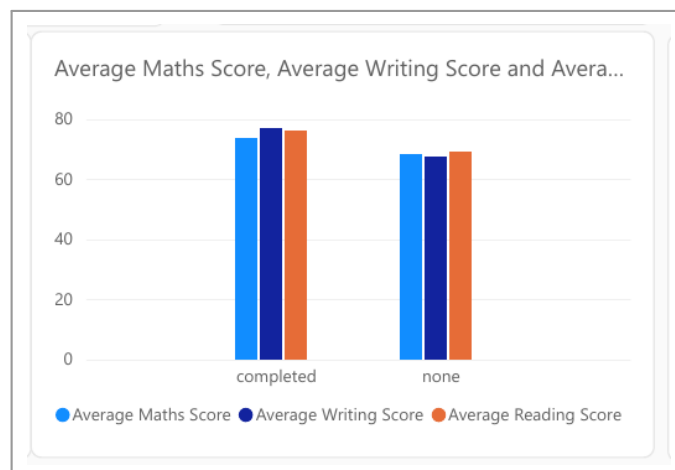
2.3 Average Score by Parental Education



This is a pie chart showing the Overall Average Score by parental level of education. Each of the six slices stands for one education level: master's degree, bachelor's degree, associate's degree, some college, some high school and high school. The legend identifies them and the labels give the average score and its share of the pie.

The slices are close in size, with labels such as 78, 71.7 and 66.2, which means students' scores are fairly similar across parental education levels. The highest average (about 78) appears to belong to the master's degree group, while the lowest (about 66) belongs to a group with less formal education. This suggests that higher parental education is linked to better performance, although the difference is modest.

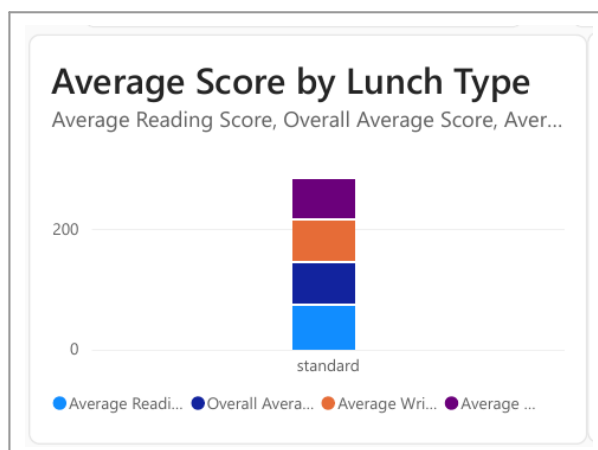
2.4 Maths, Writing and Reading Scores by Test Preparation Course



This is a clustered column chart that compares the Average Maths Score, Average Writing Score and Average Reading Score for two groups: students who completed the test preparation course and students who did not (none).

The students who completed the course score clearly higher in all three subjects, at roughly 74 to 77 marks, against about 68 to 69 for those who did not. This is the clearest result on the dashboard and shows that the test preparation course helps improve performance.

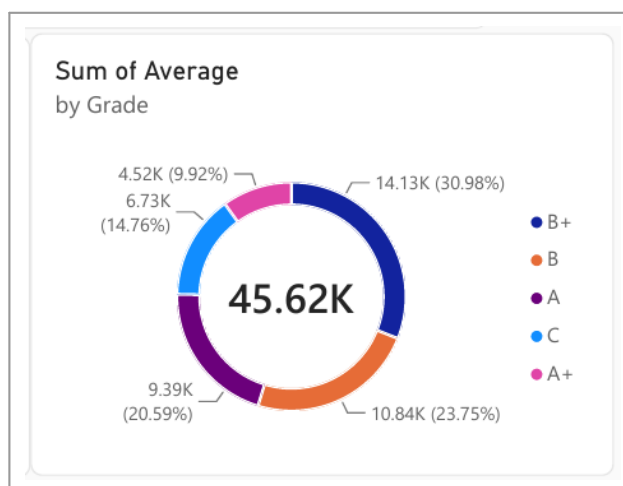
2.5 Average Score by Lunch Type



This is a stacked column chart of Average Reading Score, Overall Average Score, Average Writing Score and Average Maths Score by lunch type. Each coloured block in the column is one of these four measures, and the legend identifies the colours.

Only the standard lunch type is displayed because the lunch type slicer has standard selected. The stacked column is tall (about 280 on the axis) because four averages are added together. The chart shows how the subject scores contribute to the total for standard lunch students. Selecting free/reduced in the slicer would show the other group for comparison.

2.6 Sum of Average by Grade



This is a donut chart showing the Sum of Average by Grade, with the total of 45.62K shown in the centre. It splits the students into grade bands: B+ (14.13K, 30.98%), B (10.84K, 23.75%), A (9.39K, 20.59%), C (6.73K, 14.76%) and A+ (4.52K, 9.92%).

The B+ grade forms the largest share at about 31%, followed by B at about 24%. Grades A and A+ together make up about 30%, which shows that a good number of students perform at the top level. Grade C is only about 15%, which fits the high pass percentage seen in the KPI cards.

2.7 Slicers: Race/Ethnicity, Parental Level of Education and Lunch Type

The image shows three interactive slicers arranged horizontally. The first slicer, 'race/ethnicity', has five unchecked checkboxes for 'group A', 'group B', 'group C', 'group D', and 'group E'. The second slicer, 'parental level of educ...', has five unchecked checkboxes for 'associate's degree', 'bachelor's degree', 'high school', 'master's degree', and 'some college'. The third slicer, 'lunch type', has two buttons: 'free/reduced' (light gray) and 'standard' (dark gray, indicating it is selected).

The bottom of the dashboard has three slicers that make the report interactive.

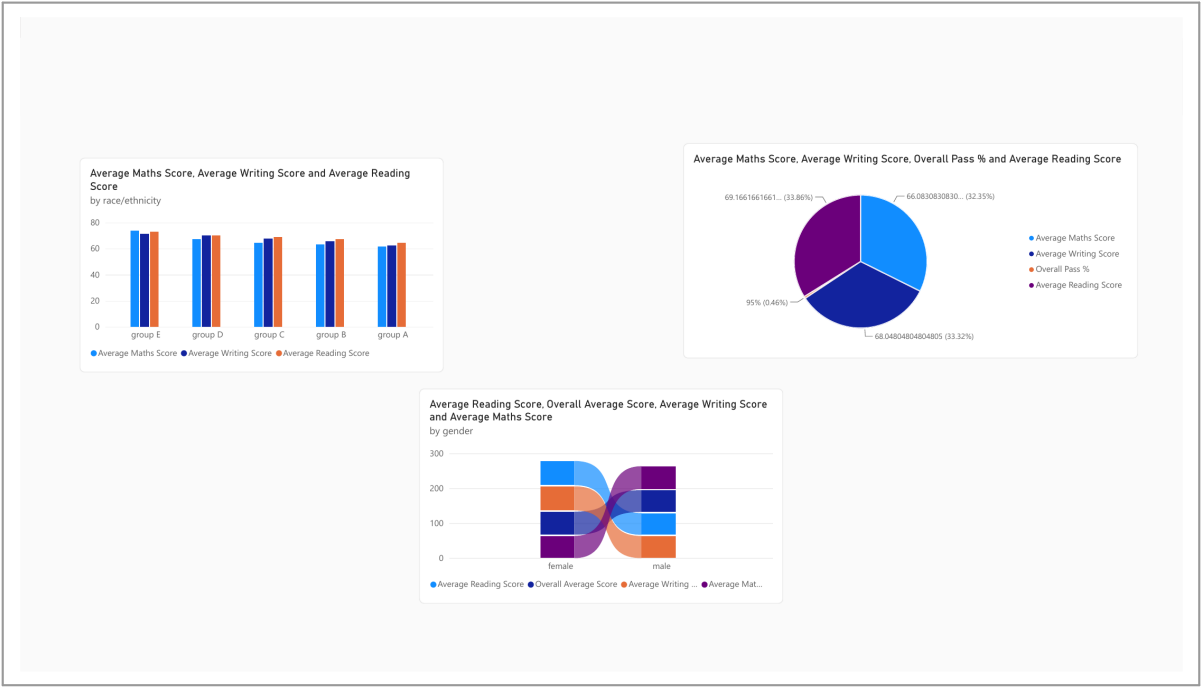
race/ethnicity: a checkbox list with group A, group B, group C, group D and group E. Ticking one or more groups filters every visual to those students.

parental level of education: a checkbox list with associate's degree, bachelor's degree, high school, master's degree, some college and some high school.

lunch type: two buttons, free/reduced and standard. The standard button is shown dark, which means it is currently selected. This is why the lunch type chart shows only the standard group.

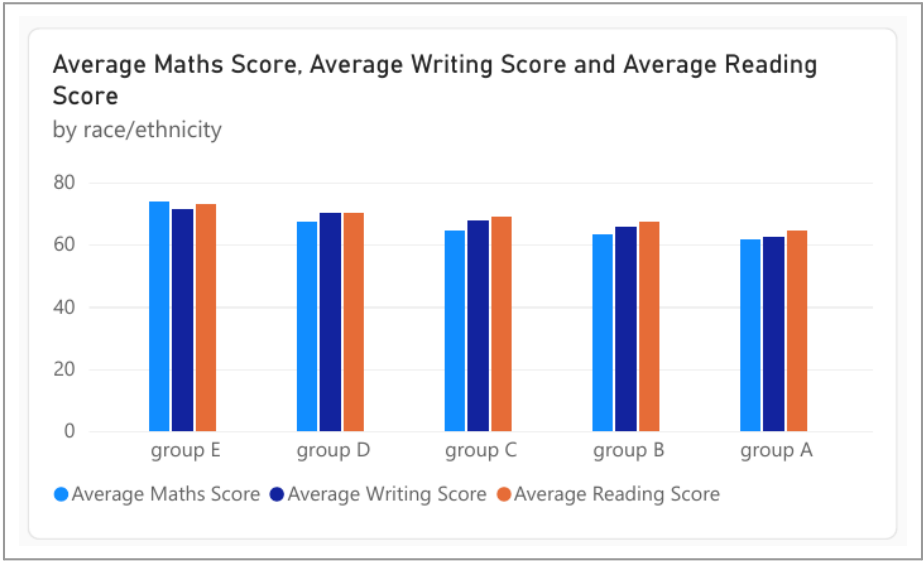
Together the slicers let the reader explore the data for any combination of group, parental education and lunch type, and the KPI cards and charts all update at once.

3. Additional Analysis Page



The third report page contains three further visuals that examine the scores by race/ethnicity, by average measure and by gender. The next pages explain each one.

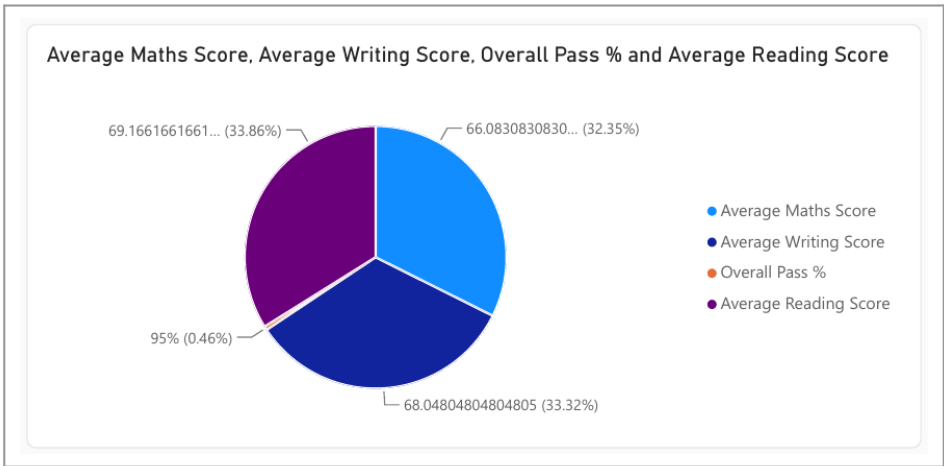
3.1 Scores by Race/Ethnicity



This is a clustered column chart showing the Average Maths Score, Average Writing Score and Average Reading Score for each race/ethnicity group, from group E to group A.

Scores fall steadily from group E to group A. Group E has the highest averages, at roughly 72 to 74 in every subject, while group A has the lowest, at roughly 62 to 65. Within most groups reading and writing are slightly above maths, with group E as the exception where maths is highest. The chart shows that performance differs between groups and that there is a gap of around 10 marks between the top and bottom groups.

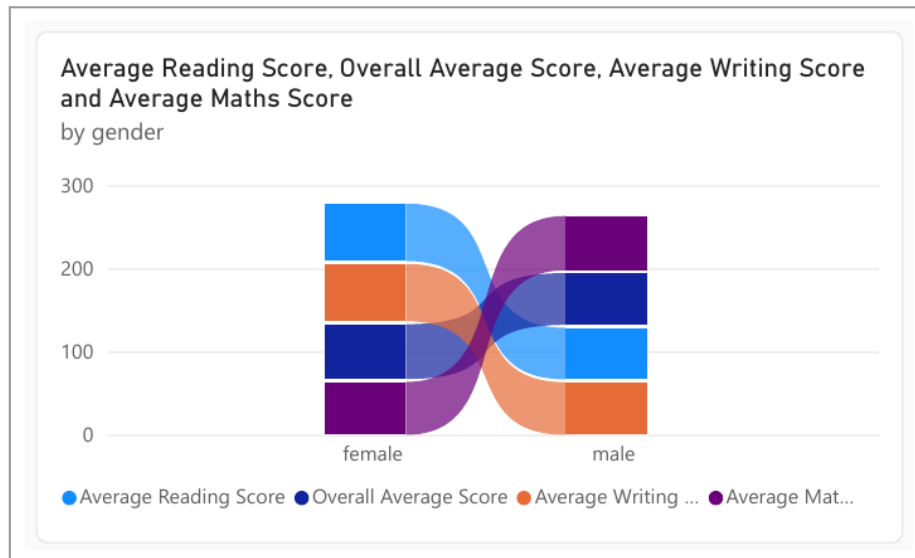
3.2 Average Scores and Pass % (Pie Chart)



This pie chart plots the Average Maths Score (66.08, 32.35%), Average Writing Score (68.05, 33.32%), Overall Pass % (95%, 0.46%) and Average Reading Score (69.17, 33.86%).

The three subject averages take almost equal slices of about one third each, with reading the largest and maths the smallest. The pass percentage appears as a very thin slice because 95 is a different kind of value from the scores. A pie chart is not the ideal choice for mixing marks and percentages, so the Pass % is better kept in a card, as on the KPI page. The values here match those of the KPI Summary page.

3.3 Scores by Gender (Ribbon Chart)



This is a ribbon chart showing the Average Reading Score, Overall Average Score, Average Writing Score and Average Maths Score by gender, with female and male as the two categories.

Each column is divided into blocks for the four measures, and the ribbons between the columns show how the ranking of the measures changes from female to male. The female column is slightly taller (about 280) than the male column (about 265), which means female students score a little higher on average. The crossing ribbons show that the order of the subjects differs between the two genders.

4. Conclusion

The Week 8 project built a set of Power BI reports to analyse student performance: a KPI summary page, an interactive analytics dashboard with slicers, and a page of additional comparisons.

The KPI cards show strong overall results, with an average score in the high 60s to low 70s and a pass percentage between 95% and 98%. Reading is the strongest subject and Maths the weakest in every view. The analysis also shows that students who completed the test preparation course score clearly higher, that scores differ across race/ethnicity groups, and that parental education and gender have a smaller but visible effect. Most students fall in grades B+ and B, with about 30% in grades A and A+.

The project demonstrates the main Power BI skills of data modelling, DAX measures, choosing suitable visuals and using slicers to make a report interactive. The insights can help in planning support for weaker subjects and in promoting test preparation courses.